Xinyi Xie

Email: sinyi.xie@gmail.com | Tel: +86 15858023209

State Key Laboratory of Advanced Materials for Intelligent Sensing, China

EDUCATION

Tianjin University Tianjin, China

- M.S. in Chemistry Sep. 2024 − Jun. 2027
- **Advisor**: Prof. Liqiang Li; Prof. Xiaosong Chen
- **GPA**: 3.71/4.0

Wuhan Institute of Technology Wuhan, China

- B.Eng. in Energy and Power Engineering Sep. 2020 − Jun. 2024
- **GPA**: 3.75/4.0, Rank: 3/131

PUBLICATIONS

[1] **Xinyi Xie**, Jie Zhu, Shougang Sun, Xiaoman Zeng, Ye Xin, Xiaosong Chen*, Yajing Sun* and Liqiang Li*, Self-Limited Growth of Centimeter-Scale Monolayer Molecular Crystals via Physical Vapor Transport. (under review)

[2] Qian Zhang#, Yongxu Hu#, Jiaxin Yan, Hengyue Zhang, **Xinyi Xie**, Jie Zhu, Huchao Li, Xinxin Niu, Liqiang Li, Yajing Sun*, Wenping Hu*, Large-Language-Model-Based AI Agent for Organic Semiconductor Device Research. *Advanced Materials*, **2024**, *36*, 2405163.

[3] **Xinyi Xie**, Lingen Chen*, Yong Yin, Shuangshuang Shi, Multi-Objective Optimization for Quantum Rectangular Cycle with Power, Efficiency and Efficient Power. *Acta Physica Polonica: A*, **2024**, *145*, 16-27.

RESEARCH EXPERIENCE

Self-Limited Growth of Centimeter-Scale Monolayer Molecular Crystals via Physical Vapor Transport

Primary Contributor and First Author

This work proposed a self-limited growth strategy for directly growing centimeter-scale organic semiconductor monolayer molecular crystals (MMCs) on SiO_2/Si substrates via PVT. The method avoided transfer-induced defects associated with solution processing and the dependence of van der Waals epitaxy on two-dimensional templates.

- Introduced H_2 to grow C_{10}-DNTT MMCs with dimensions up to 8.2 cm × 2 cm and confirmed large-area uniformity.
- Fabricated OFET arrays that showed an average mobility of 7.19 $cm^2\ V^{-1}\ s^{-1}$ and a contact resistance of 37.3 Ω·cm.
- Used MD simulations to reveal that self-limited growth of MMCs on amorphous substrates arises from the synergistic effect of molecular side chains and substrate surface energy.
- Achieved patterned MMC growth by photolithographically modulating the substrate surface energy.
- Extended the method to p-type and n-type OSCs and further achieved vertical and lateral heterojunction growth.

Wafer-Scale Self-Limited Growth of Monolayer Molecular Crystals via Vacuum Evaporation

Primary Contributor and First Author

Building on previous work, this work further developed an improved vacuum-evaporation method to directly grow continuous and uniform 4-inch wafer-scale MMCs on SiO_2/Si substrates within 10 min. It broke a bottleneck of growing wafer-scale MMCs on amorphous substrates and provided a route toward large-scale electronic integration.

- Independently carried out hands-on modification of the vacuum-evaporation chamber, built the growth apparatus, and continuously optimized and improved both the apparatus and the method.
- Fabricated wafer-scale, high-density OFET arrays with high mobility and excellent device uniformity, and further explored short-channel device fabrication.
- Used photolithography to pattern electrode and gate structures, then fabricated wafer-scale top-gate inverter arrays through transfer-based integration to explore circuit-level applications.

Few-Layer Covalent Fullerene Networks via PECVD

Primary Contributor and Co-first Author (listed second)

This work developed a PECVD method to grow bulk crystalline C_{60} polymer (graphullerite) that could be mechanically exfoliated into molecularly thin graphullerene sheets with clean interfaces, with potential for 2D electronic devices.

- Optimized PECVD growth conditions to achieve controlled growth of millimeter-scale graphullerite.
- Mechanically exfoliated graphullerite into few-layer graphullerene sheets, achieving a minimum thickness of 1.5 nm.
- Characterized the material properties using Raman, PL, SEM, TEM, AFM-IR, and XPS.
- Measured thermal conductivity using TDTR and SThM, revealing that thermal conductivity increased with decreasing thickness, with up to a tenfold difference between bulk and few-layer samples.

Large-Language-Model-Based AI Agent for Organic Semiconductor Device Research

Contributing Member

This work developed an LLM-based AI agent automating OFET literature data extraction and integrating machine learning for predictive modeling and data-guided optimization, achieving a threefold increase in OFET mobility.

- Constructed a standardized OFET database from 277 articles, with 709 device configurations and over 10,000 records.
- The systematic evaluation of GPT-4 across 14 parameter categories yielded average precision and recall both above 92%, demonstrating the reliability of LLMs for scientific information extraction.
- Benchmarking five machine learning algorithms including XGBoost and Random Forest identified XGBoost as the optimal classifier (F1 = 0.88). DFT results confirmed that optimization strategies increased DP-DTT OFET mobility.

PATENT

[1] Yajing Sun (Advisor), **Xinyi Xie**, Shougang Sun, Xiaosong Chen, Liqiang Li, Wenping Hu, “A Method for Growing Organic Semiconductor Monolayer Films on Amorphous Substrates”, Chinese Patent, CN120925072B, 2025-01-30.

AWARDS & HONORS

Awards:

- **National First Prize**, China Undergraduate Physics Experiment Competition (CUPEC) 2022
- **National First Prize**, National College Students’ Energy Conservation and Emission Reduction Competition 2023
- **National Second Prize**, Chinese Undergraduate Mathematics Competition 2023
- **Hubei Provincial First Prize**, Lan Qiao Cup National Software and Information Technology Professionals Competition (Python Category) 2023

Honors:

- The First Prize Graduate Scholarship, Tianjin University 2024 − 2026
- The First Prize Scholarship, Wuhan Institute of Technology 2020 − 2024
- **National** Top 100 Outstanding Undergraduate Theses (Design Projects) in Energy and Power Engineering 2024